Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/13/2005
PI Name:	Jin, J.-P. M.D., Ph.D.		
Project Title:	Adaptation Of Myocardial Contractile Proteins In Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	jpjin@northwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	847.570.1960
Organization Name:	Evanston Northwestern Healthcare Research Institute		
PI Address 1:	2650 Ridge Ave		
PI Address 2:	Molecular Cardiology Division		
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60201-1718	**Congressional District:**	9
Comments:	Former affiliation: Case Western Reserve University; moved in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK62G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In a rat tail-suspension model of simulated microgravity, we found an up-regulated proteolytic modification of cardiac troponin I (cTnI), an essential protein in the Ca2+-regulatory system of cardiac muscle. This novel structural regulation of cTnI removes the N-terminal 26-27 amino acids, including Ser23 and Ser24 that are substrates of cAMP-dependent protein kinase (PKA). Beta-adrenergic-regulated PKA phosphorylation of cTnI reduces the sensitivity of cardiac muscle to Ca2+. Therefore, the up-regulation of N-terminal truncated cTnI may provide a compensation for cardiac function against the microgravity-induced reduction of contractility. This research proposal describes ground based studies to investigate the role of the proteolytic regulation of cTnI in adaptation to microgravity for potential use in the prevention and treatment of cardiovascular deconditioning during and after space flight. We plan to study the functional changes in cTnI caused by the N-terminal truncation and the effect on cardiac muscle contraction in simulated microgravity. Conformation and function of the N-terminal truncated cTnI will be investigated in vitro and in the heart of transgenic		

	mice. We shall examine the effect of the removal of PKA phosphorylation sites from cTnI on beta-adrenergic regulation during cardiac adaptation in simulated microgravity. We shall also investigate the regulatory mechanisms for this proteolytic modification of cTnI during adaptation to microgravity. We shall focus on the role of calpain that is a family of regulatory proteases and known to function in the degradation of troponin subunits. The role of PKA and PKC phosphorylation of cTnI in the regulation of sensitivity to calpain cleavage will be characterized. This research will provide new information for understanding cardiac adaptation in microgravity and protecting cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our study will contribute to both space life science and human life quality on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Progress during the period of June 1, 2004 to May 31, 2005 1) Characterization of the function of N-terminal truncated cardiac troponin I: To investigate the role of the N-terminal truncated cardiac troponin I in myocardial adaptation to microgravity, we examined the cardiac function of transgenic mice that we developed to over-express the N-terminal truncated cardiac troponin I (cTnI-ND) in the heart. Ca2+-activated actomyosin ATPase activity showed that cTnI-ND myofibrils had lower affinity for Ca2+ than controls, similar to the effect of isoproterenol treatment. In vivo and isolated working heart experiments revealed that cTnI-ND hearts had a significantly faster rate of relaxation and lower left ventricular end diastolic pressure (LVEDP) compared to controls. The higher baseline relaxation rate of cTnI-ND hearts was at the level of b-adrenergic stimulated wild type mouse hearts. The decrease in cardiac output due to lowered preload was significantly smaller for the cTnI-ND hearts compared to controls. These findings indicate that removal of the N-terminal extension of cTnI via restricted proteolysis enhances cardiac function by increasing the rate of myocardial relaxation and lowering LVEDP to facilitate ventricular filling, thus resulting in better utilization of the Frank-Starling mechanism. This is especially important when ventricular filling is decreased as that seen after long term exposure to microgravity. A comprehensive report for this work has been published: Barbato, J.C., Huang, Q.-Q., Hossain, M.M., Bond, M., and Jin, J.-P. (2005) Proteolytic N-Terminal Truncation of Cardiac Troponin I Facilitates Ventricular Relaxation and Enhances Heart Function J. Biol. Chem. 280:6602-6609. 2) Mechanisms that regulate the proteolytic N-terminal truncation of cardiac troponin I: To investigate the regulation of the cardiac troponin I N-terminal truncation and the role of phosphorylation, we have demonstrated m-calpain cleavage of cardiac troponin I in isolated myofibrils. Interestingly, post mortal dephosphorylation of troponin I seems to enhance the N-terminal truncation of cardiac troponin I, indicating that the phosphorylation states of cardiac troponin I may play a role in regulating the proteolytic N-terminal truncation. This hypothesis is currently investigated using our engineered cardiac troponin I in which Ala or Asp substitutions are made at Ser23 and Ser 45 to mimic the non-phosphorylated and phosphorylated states. 3) In addition to the development of countermeasures to the decreased ventricular filling during long space flights, our finding that N-terminal truncation of cardiac troponin I is an effective novel mechanism to enhance myocardial relaxation has an attractive application on earth as a new target for the treatment of heart failure. We have submitted a provisional US patent application for this application. 4) An interesting fact we observed is that truncation of the N-terminal extension of cardiac troponin I as seen in simulated microgravity rat models has similar effect on enhancing cardiac muscle relaxation to that of cardiac troponin I N-terminal phosphorylation upon beta—adrenergic stimulation. Therefore, various structural modifications at the N-terminal region of cardiac troponin I may have the same down stream effect. To initiate a search of chemicals that bind the N-terminal region and reproduce such effect, we have submitted an NIH Roadmap grant application (R03 MH076314, PI: J.-P. Jin).
Bibliography Type:	Description: (Last Updated: 04/20/2009)
Articles in Peer-reviewed Journals	Barbato JC, Huang QQ, Hossain MM, Bond M, Jin JP. "Proteolytic N-terminal truncation of cardiac troponin I enhances ventricular diastolic function." J Biol Chem. 2005 Feb 25;280(8):6602-9. PMID: 15611140 , Feb-2005